Stick Game

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1729 | Mathematics | https://cses.fi/problemset/task/1729

Consider a game where two players remove sticks from a heap. The players move alternately, and the player who removes the last stick wins the game.

A set $P=\{p_1,p_2,\ldots,p_k\}$ determines the allowed moves. For example, if $P=\{1,3,4\}$, a player may remove 1, 3 or 4 sticks.

Your task is find out for each number of sticks $1,2,\ldots,n$ if the first player has a winning or losing position.

Input

The first input line has two integers n and k: the number of sticks and moves.

The next line has k integers $p_1,p_2,\ldots,p_k$ that describe the allowed moves. All integers are distinct, and one of them is 1.

Output

Print a string containing n characters: `W` means a winning position, and `L` means a losing position.

Constraints

- $1 \le n \le 10^6$
- $1 \le k \le 100$
- $1 \le p_i \le n$

Example

Input

```
10 3
1 3 4
```

Output

```
WLWWWWLWLW
```